

Recent Advances In AEC and NS

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 **声学回声消除技术最新进展**

 **噪声抑制技术最新进展**



Acoustic Echo Cancellation & Deep Noise Suppression Challenge

[About](#) [Rules](#) [Timeline](#) [Organizers](#) [Links](#) [Results](#)

Program dates: December 2021-February 2022

The [ICASSP 2022](#) Acoustic Echo Cancellation Challenge is intended to stimulate research in the area of acoustic echo cancellation (AEC), which is an important part of speech enhancement and still a top issue in audio communication and conferencing systems. This is the third AEC challenge and it is enhanced by including mobile scenarios, adding speech recognition rate in the challenge goal metrics, and making the default sample rate 48 kHz. In this challenge, we open source two large datasets to train AEC models under both single talk and double talk scenarios. These datasets consist of recordings from more than 7,500 real audio devices and human speakers in real environments, as well as a synthetic dataset. We also open source an online subjective test framework and provide an online objective metric service for researchers to quickly test their results. The winners of this challenge are selected based on the average Mean Opinion Score achieved across all different single talk and double talk scenarios, and the speech recognition rate.

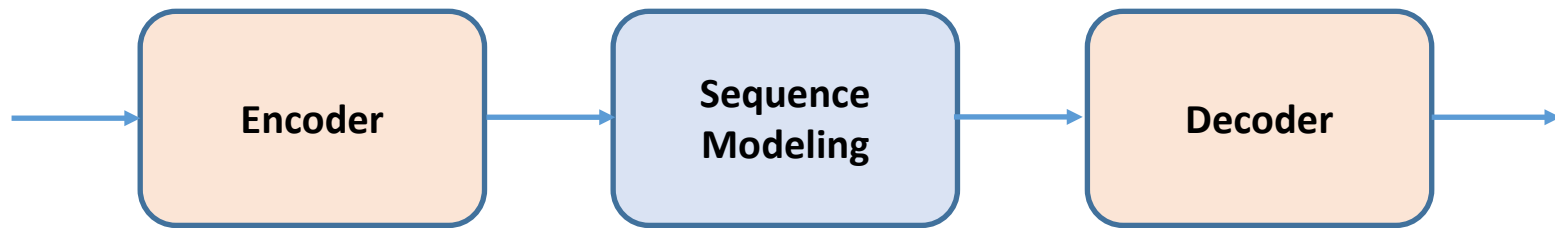
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Noise suppression has become more important than ever before due to the increasing use of voice interfaces for various applications. Given the millions of internet-connected devices being employed for audio/video calls, noise suppression is expected to be effective for all noise types chosen from daily-life scenarios. The IEEE ICASSP 2022 Grand Challenge is the 4th DNS challenge intended to promote industry-academia collaboration on research in real-time noise suppression aimed to maximize the subjective (perceptual) quality of enhanced speech. This challenge will extend DNS efforts to full band speech with a special focus on personalized denoising. In the era of hybrid work, personalized denoising is very important to suppress neighboring speaker and/or background noises. Recently, DNS research has been moving fast, and researchers now have state-of-the-art advancements in deep neural networks (DNNs); currently, deep noise suppression methods leverage the convolutional, recurrent, or hybrid neural network for estimating the enhanced speech from noisy recordings.

<https://www.microsoft.com/en-us/research/academic-program/deep-noise-suppression-challenge-interspeech-2020/>



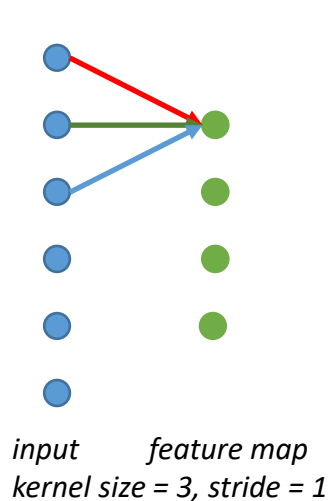
Encoder – Sequence Modeling – Decoder 架构



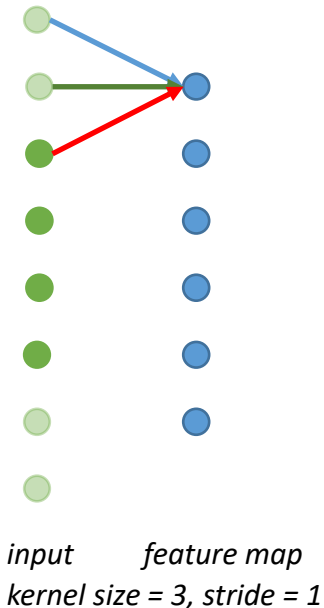
- 最初，编码器/解码器通常由Convolutional Network组成，序列建模部分由通常由Recurrent Network组成。
- 因此，这种架构也经常被称为Convolutional Recurrent Network (CRN)、Convolutional Encoder/Decoder (CED) with Recurrent Network。
- 绝大多数基于深度学习的语音增强、回声消除、语音分离方法，都可以看做是在这个基础架构上的扩展。



Convolution & Deconvolution (Transposed Convolution)



Conv1D



TrConv1D

- 任何一个convolutional layer, 都可以找到一个transposed convolutional layer与之对应。

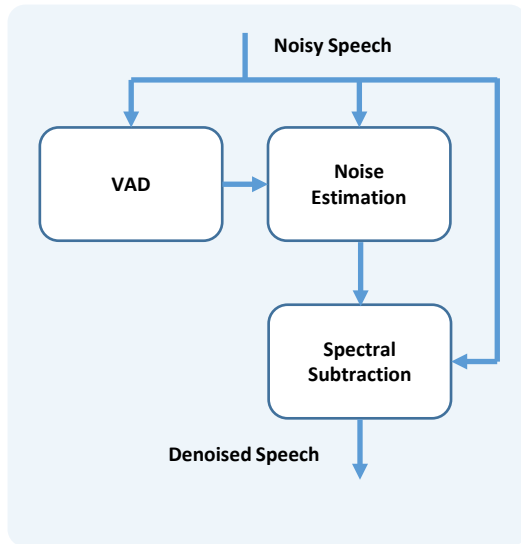
Transposed convolution = full convolution + weight transpose

- 所以, 之后对各种方法的解读, 我们只重点关注Encoder和Sequence Modeling部分。

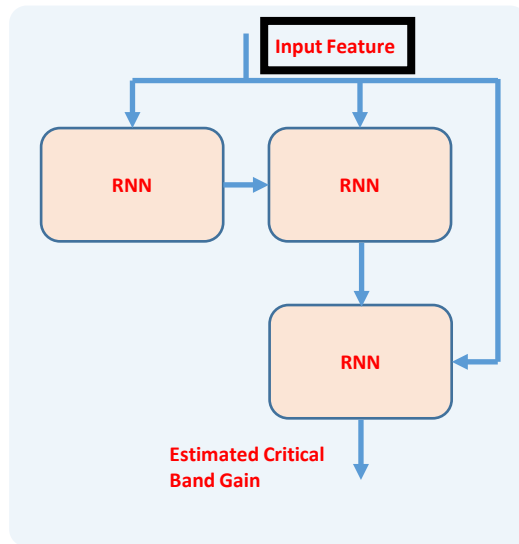


AEC--从RNNoise说起

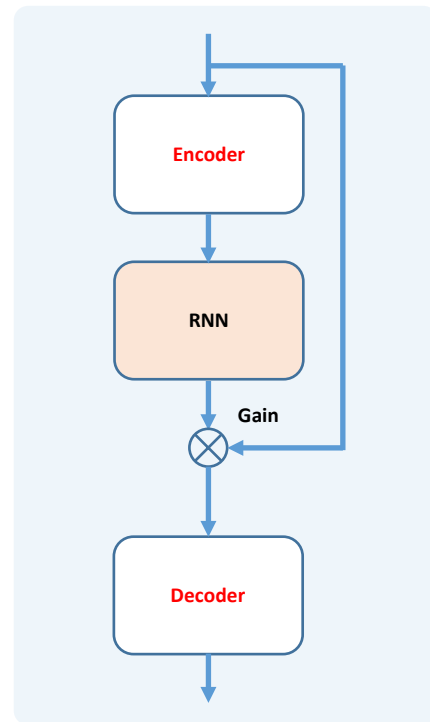
Noise Suppression based on DSP



Neural Network for NS



RNNoise – A hybrid structure



- **VAD**: Speech absence? When to update noise estimation?
- **Noise Estimation**: What to suppress?
- **Spectral Subtraction**: How to suppress?



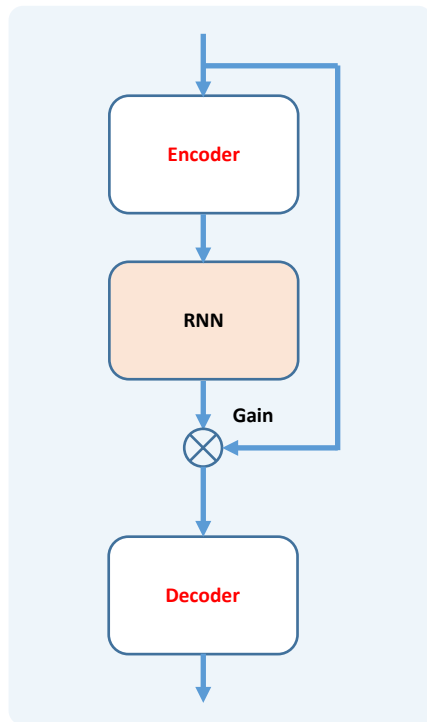
RNNoise

8k ~ narrow band

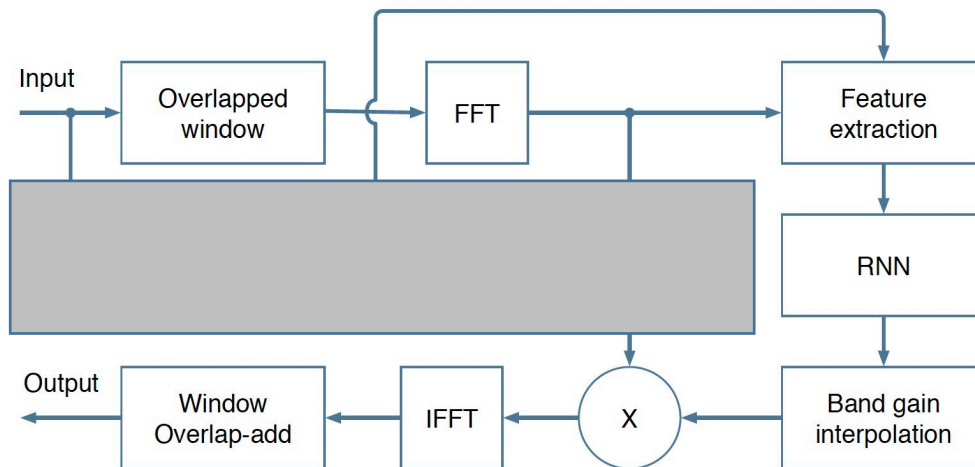
16k ~ wideband

46k ~ full-band

RNNoise



RNNoise block diagram in detail:

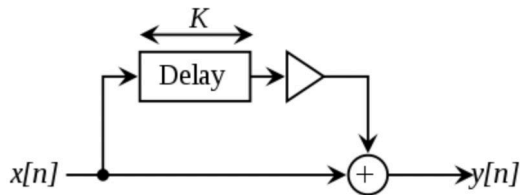


- *Feature Extraction*: 42-dim, including Bark-frequency cepstral coefficients BFCC, its derivatives, pitch information of *full-band* speech, etc.
- *Critical Band Gain*: 22 values in the [0,1] range, interpolated to each frequency bin.
- *Pitch filtering*: 22 Bark bands are TOO COARSE. To model finer details in the spectrum, *pitch comb filter* is used to suppress noise between pitch harmonics.



何为“梳状滤波器”

梳状滤波器 (comb filter)通过向自身添加信号延迟而实现, 具备削弱干扰, 增强周期性信号的功能。



FIR Comb filter

$$y(n) = x(n) + \alpha x(n - K)$$

$$Y(k) = X(k) + \alpha e^{-j\omega K} X(k)$$

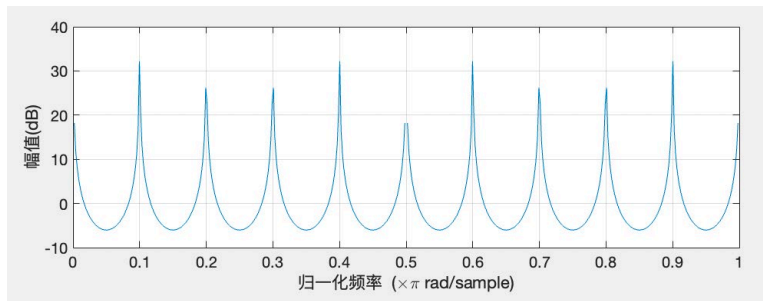
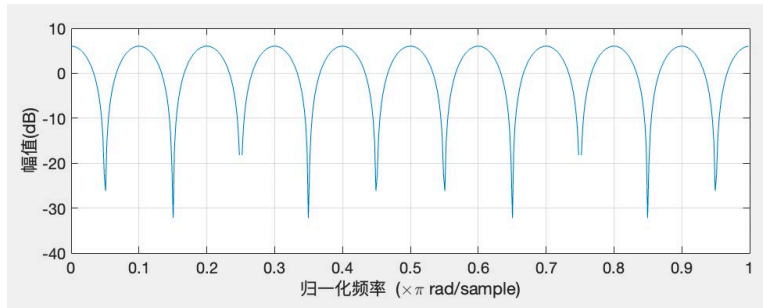
$$\Rightarrow H(e^{j\omega}) = 1 + \alpha e^{-j\omega K}$$

IIR Comb filter

$$y(n) = x(n) + \beta y(n - K)$$

$$Y(k) = X(k) + \beta e^{-j\omega K} Y(k)$$

$$\Rightarrow H(e^{j\omega}) = \frac{1}{1 - \beta e^{-j\omega K}}$$

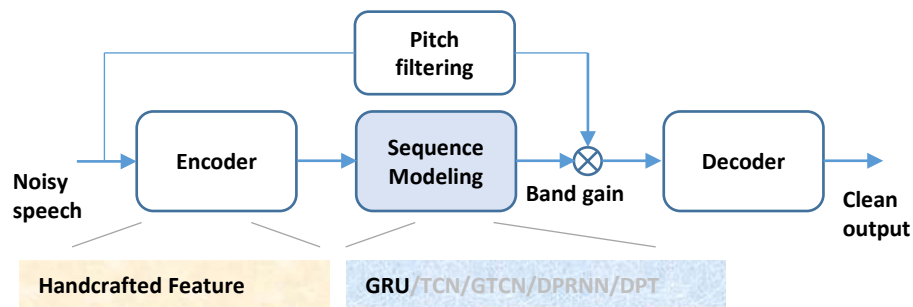


$$K = 20, \alpha = 1, \beta = 1$$



RNNoise

RNNoise



RNNoise的性能

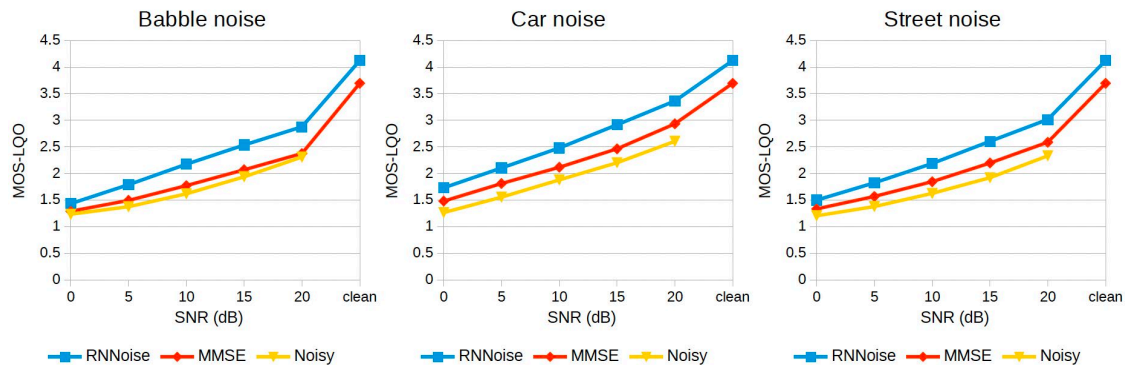
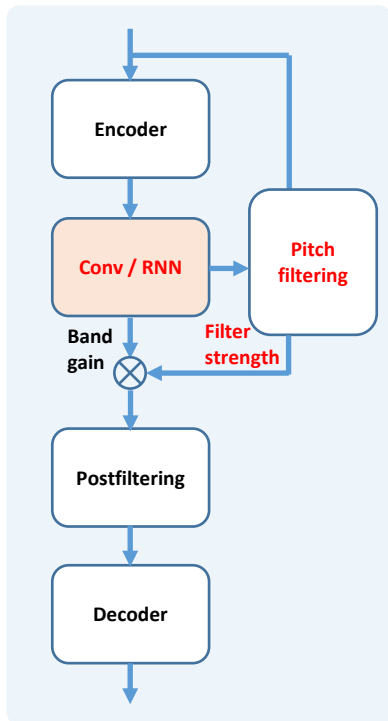


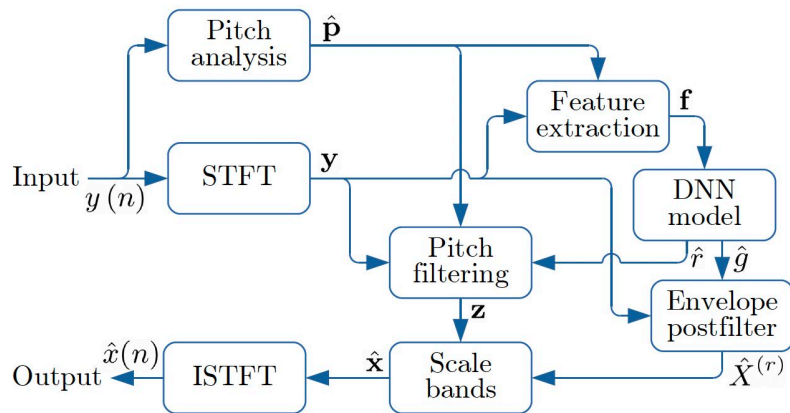
Figure 5. PESQ MOS-LQO quality evaluation for babble, car, and street noise.



PercepNet



PercepNet diagram in detail:



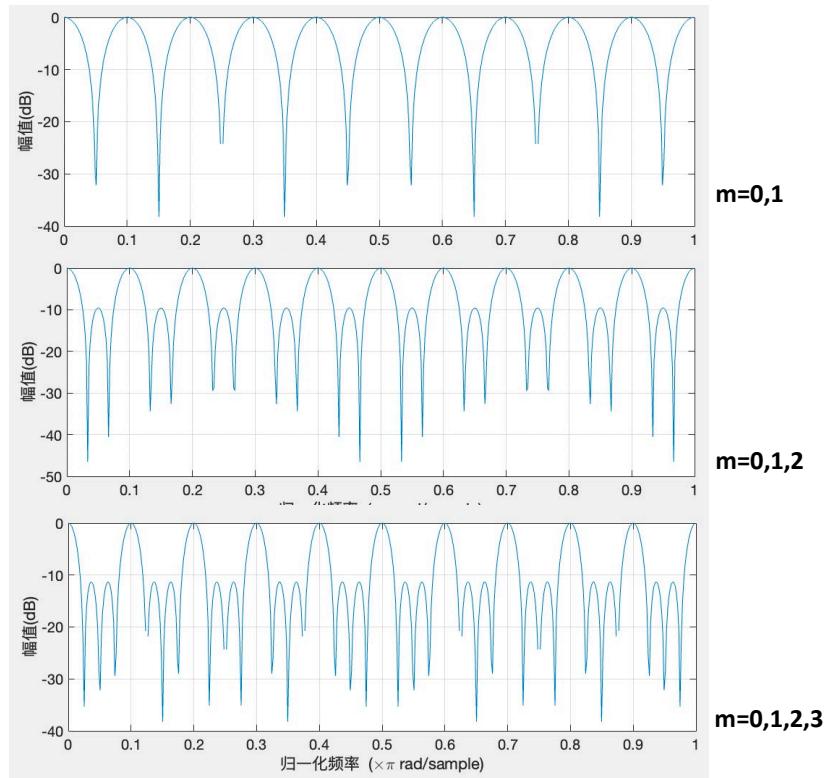
Difference with RNNoise:

- *DNN input*: 70-dim, including 34-band magnitude and pitch information.
- *DNN output*: 34 comb filter parameters are also learned by DNN, besides 34 band gains.
- *Pitch filtering*: **multi-tap comb filter**.
- *DNN structure*
- *Postfiltering*

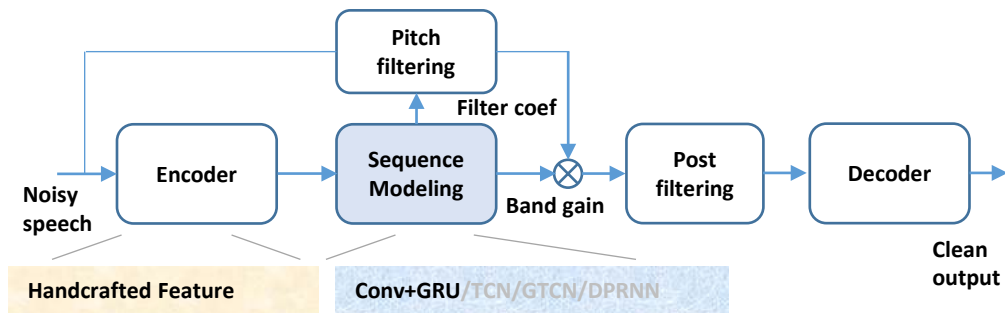


Multi-tap Comb filter

$$H(e^{j\omega}) = \sum_{m=-M}^M w_m e^{-j\omega mK}$$



PercepNet:



PercepNet性能:

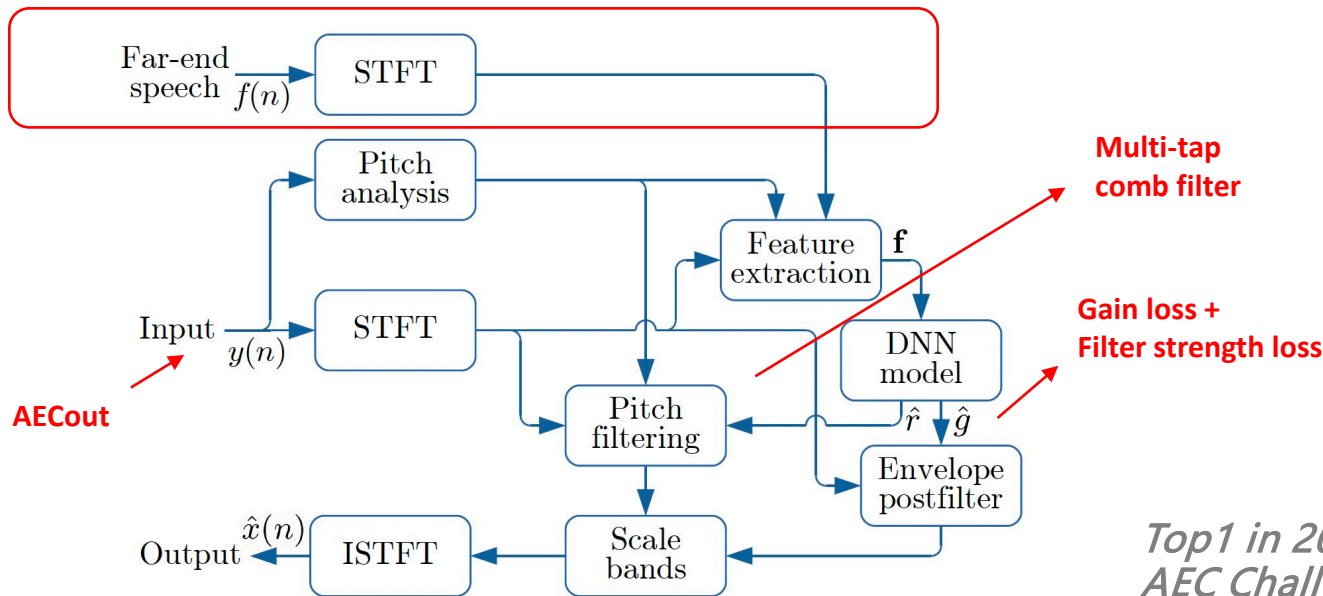
Table 1: *P.808 MOS results based on internal testing on the VCTK test set at 48 kHz.*

Algorithm	PESQ-WB	MOS (P.808)
Noisy	1.97	3.40
SEGAN [9]	2.16	-
RNNoise (original) [12]	2.29	3.70
PercepNet (no pitch, no pf)	2.64	3.81
PercepNet (no pf)	2.73	3.91
PercepNet (no pitch)	2.47	3.93
PercepNet	2.54	4.05



PercepNet-AEC

PercepNet-AEC架构:

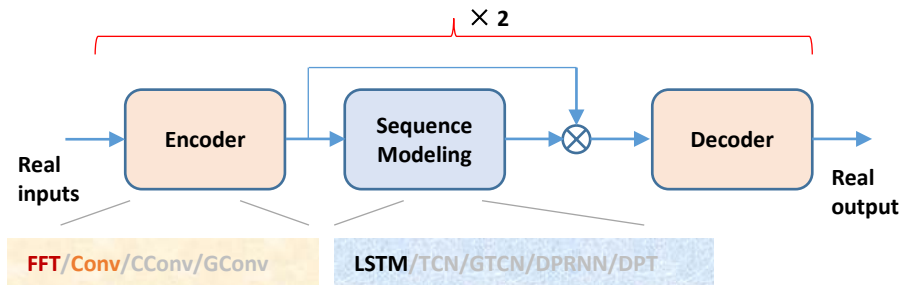


*Top1 in 2021 ICASSP
AEC Challenge*

Low-Complexity, Real-Time Joint Neural Echo Control and Speech Enhancement based on PercepNet. J.M Valin, 2021

Dual-Signal Transformation LSTM Network

DTLN-AEC:



Stacked architecture:

第一阶段:

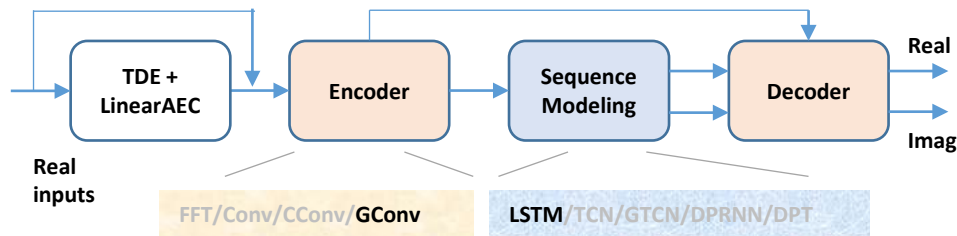
- 时频域网络, 预测time-frequency mask;
- 只考虑幅度信息, 不考虑相位信息。

第二阶段:

- 时域网络, 预测latent space mask;
- 同时考虑幅度和相位。

Gate Complex Convolutional Recurrent Network

GCCRN:

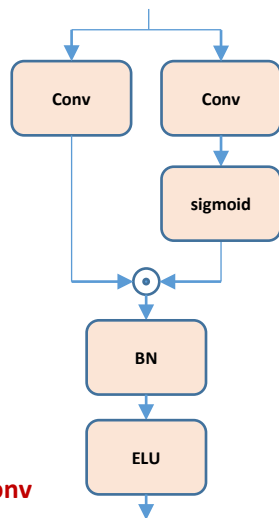
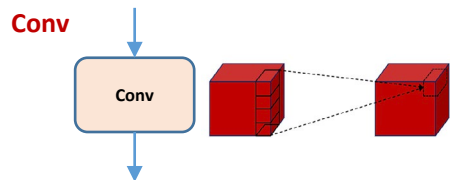


- 线性AEC + NN残余回声抑制
- 一个encoder, 两个decoder, 分别预测实部和虚部
- Gated Convolution

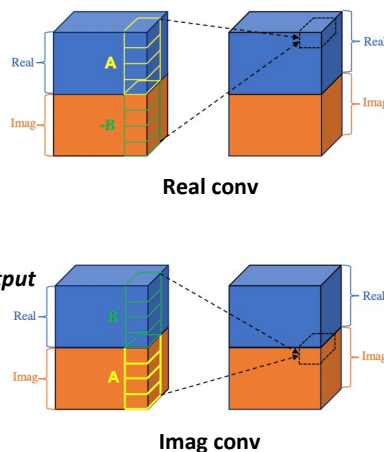
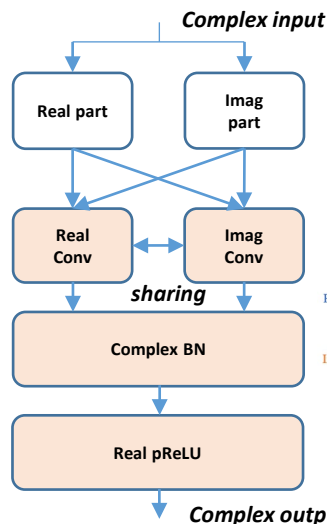


几种不同的Convolution

Conv / Gate Conv / Complex Conv :



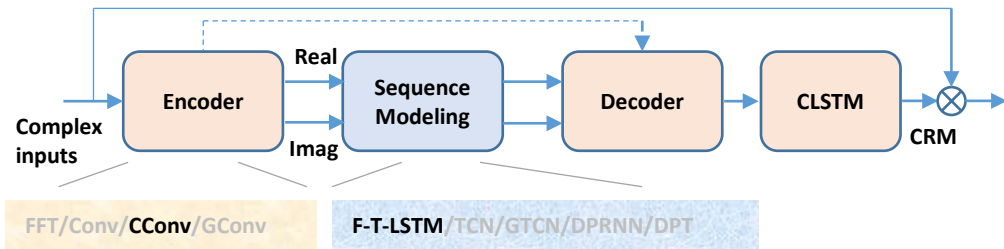
Gate Conv



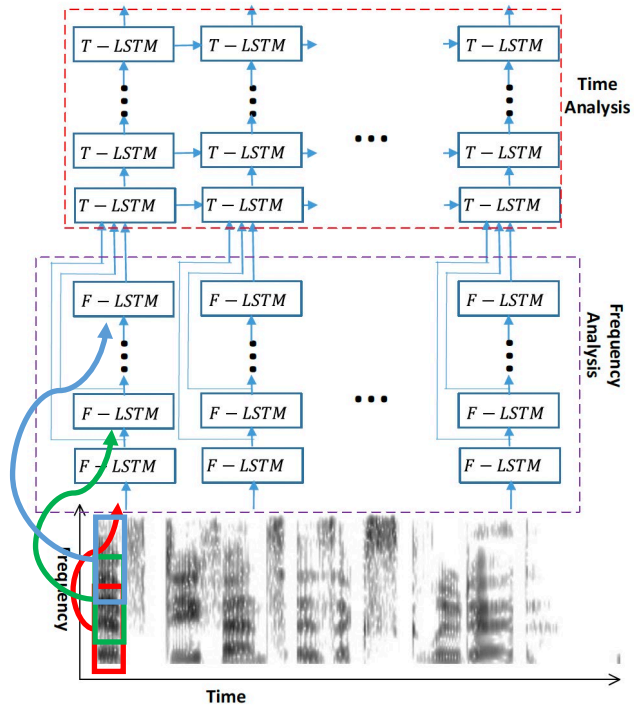
- Gate Conv: 两次卷积, 两倍运算。
- Complex Conv: 两次卷积, 四倍运算。
- Complex Conv的参数量是Real Conv的两倍。



F-T-LSTM:



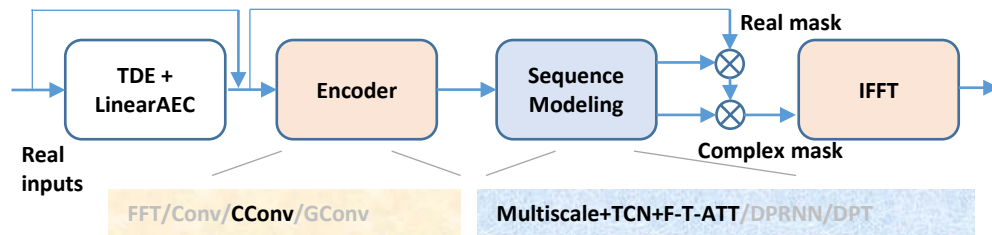
- F-biLSTM
- T-LSTM



F-T-LSTM based Complex Network for Joint Acoustic Echo Cancellation and Speech Enhancement. S Zhang, etc., 2021

Multi-scale Temporal Frequency Convolutional Network with Axial Attention

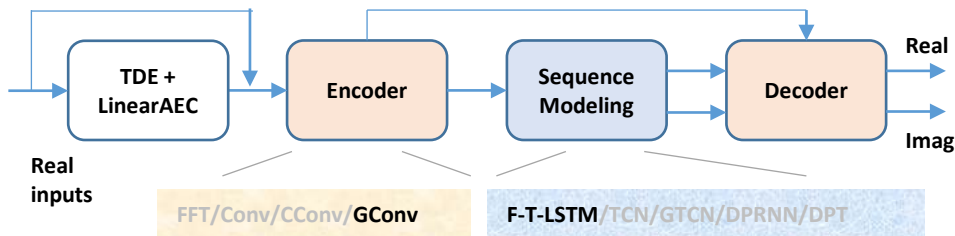
MTFAA-Net:



- 线性AEC + NN残余回声抑制
- 多尺度特征
- Temporal Frequency Model:
 - 序列建模: TCN with Conv2d
 - Axial Attention = F-T-Attention
- 两阶段的mask估计

Gated Convolutional F-T-LSTM Neural Network

GFTNN:

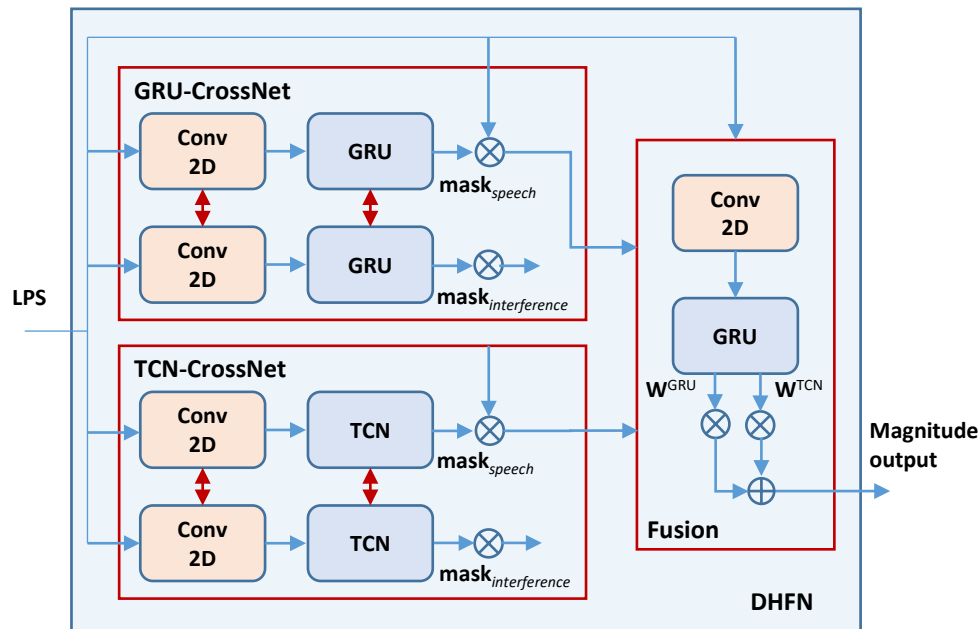
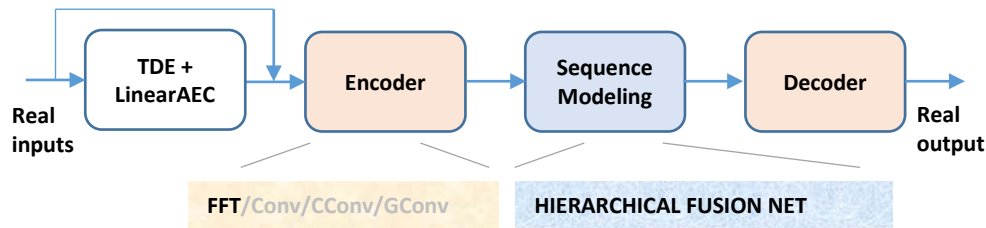


- 线性AEC + NN残余回声抑制
- 一个encoder, 两个decoder, 分别预测实部和虚部
- GCCRN with F-T-LSTM



Deep Hierarchical Fusion Network

DHFN:

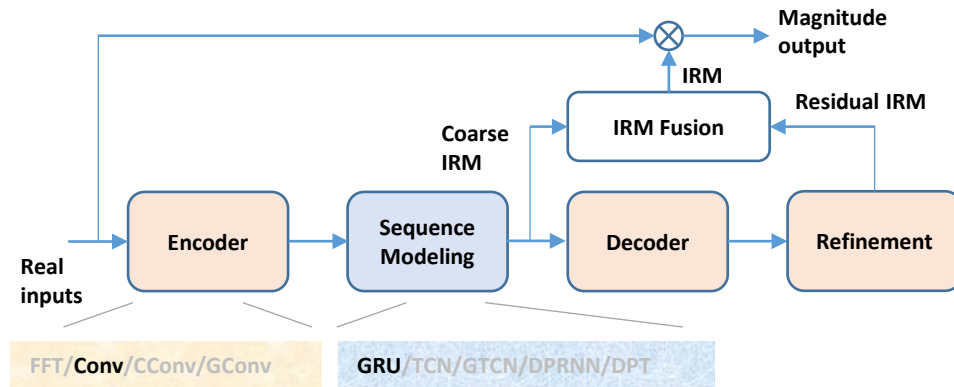


- 线性AEC + NN残余回声抑制
- Inter-network fusion
- Intra-network fusion
- 引入ASR的信息, 多阶段训练



Multi-Scale Refinement Network

MSRN:

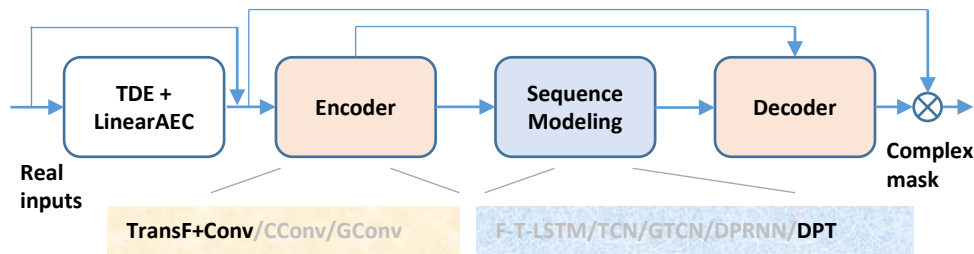


- Encoder阶段: 估计coarse mask
- Decoder阶段: 估计residual mask -- refinement



Transformer-based

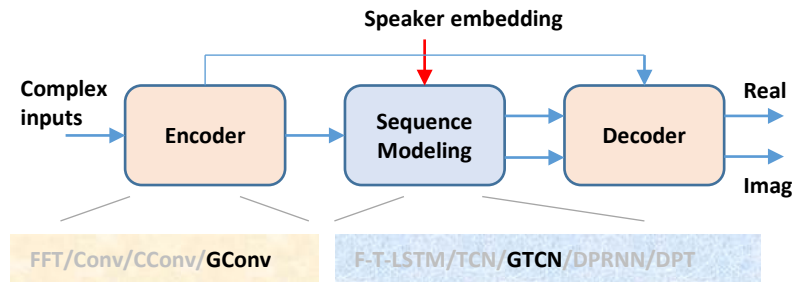
Transformer-based network:



- 线性AEC + NN残余回声抑制
- Encoder中加入了transformer, 用ref对mic做attention
- 序列建模中使用Dual-Path Transformer ($\approx$ F-T-ATT)

Gated Temporal Convolutional Neural Network

GTCNN:



- Personalized AEC
- Fast ResNet as Speaker embedding
- 一个encoder, 两个decoder, 分别预测实部和虚部



Summary

	Input feature	Network architecture	Loss function	Auxiliary Task	Training data	Real or Complex	Hybrid or not
RNNoise	Hand-crafted <i>42-dim, extracted from noisy input, including BFCC, its derivatives, pitch information, etc.</i>		<i>kind of MSE, on 'compressed' critical band gain</i>	Pitch filtering <i>Comb filter to suppress noise between pitch harmonics.</i>	6h speech + 4h noise	R	Y
Percep Net	Hand-crafted <i>70-dim, including magnitude and pitch coherence in each 34 bands, and pitch period, pitch correlation.</i>		Band gain loss + Filter strength loss	Pitch filtering <i>Multi-tap comb filter</i>	120h speech + 80h noise	R	Y

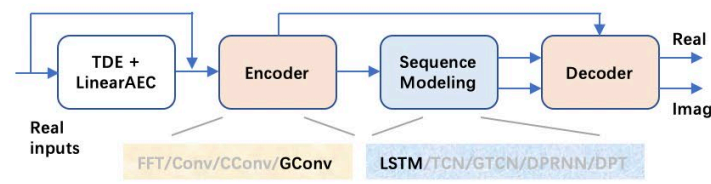
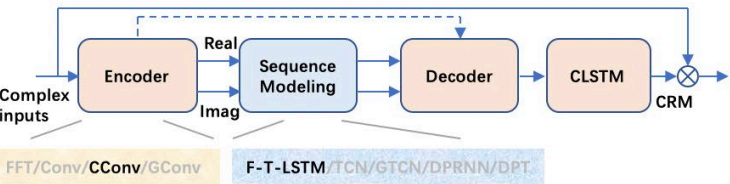


Summary

Input feature		Network architecture	Loss function	Auxiliary Task	Training data	Real or Complex	Hybrid or not
Percep-Net	Mic / Ref / AECout Time-domain signals. Feature Extraction : 100-dim, including energy (near & far-end) and pitch coherence in each 32 bands, and pitch period, pitch correlation, etc.	Real inputs AECout Encoder 32 Filter strength Pitch filtering 32 band gain Postfilter/Decoder Handcrafted Feature Conv+GRU/TCN/GTCN/DPRNN	Band gain loss + Filter strength loss	Pitch filtering Multi-tap comb filter	120h speech + 80h noise	R	Y
DTLN	Mic / Ref Time-domain signals.	Real inputs Encoder Sequence Modeling Decoder Real output FFT/Conv/CConv/GConv LSTM/TCN/GTCN/DPRNN/DPT × 2	SNR	None	60h	R	N



Summary

	Input feature	Network architecture	Loss function	Auxiliary Task	Training data	Real or Complex	Hybrid or not
GCCRN	Mic / Ref / AECout <i>Real and imaginary part of compressed complex spectrum of time-aligned ref and AECout, stacked together.</i>		MSE <i>MSE on real and imaginary part, or on magnitude and phase.</i>	None	/	C	Y
F-T-LSTM	Mic / Ref <i>Real and imaginary part stacked together.</i>		Seg-SI-SNR	None	~ 100h	C	N



Summary

Input feature	Network architecture	Loss function	Auxiliary Task	Training data	Real or Complex	Hybrid or not
MTFAA-Net AEC2022, Baidu	Mic / Ref / AECout Complex spectra input of 3 channels.	PLCPA MSE on power-law compressed spectrum, i.e. mag + phase loss.	None	/	C	Y
GFTNN AEC2022, NWPU	Mic / Ref / estEcho / AECout Different input combinations. Real and imaginary part of compressed spectrum stacked together.	Echo-aware PLCPA + VAD loss PLCPA: mag + phase loss. VAD: to punish wrong judgement of near-end.	Multi-task VAD as an auxiliary task, to punish near-end over suppression.	860h (460s + 400r)	C	Y



Summary

	Input feature	Network architecture	Loss function	Auxiliary Task	Training data	Real or Complex	Hybrid or not
DHFN AEC2022, Kuaishou	Mic / Ref / AECout <i>Log-power spectrum (LPS) of ref, mic and AECout signals concatenated together, without phase information.</i>		AEC loss + ASR loss <i>AEC loss : OSISNR + Power-law MSE, for both speech and interference.</i> <i>ASR loss : ASR embedding distance, extracted by WeNet encoder.</i>	Multi-stage training <i>S1: Train two CrossNets separately.</i> <i>S2: Add ASR loss to fine-tune two CrossNets, and fix them.</i> <i>S3: Train Fusion network</i>	>1000h	R	Y



Summary

Input feature		Network architecture	Loss function	Auxiliary Task	Training data	Real or Complex	Hybrid or not
MSRN AEC2022, Xiaomi	Mic / Ref <i>Magnitude spectrum of ref and mic signals stacked together, with phase spectrum unchanged.</i>		MSE + SI-SNR	None	/	R	N
TF-based AEC2022, Xiaomi	Ref / AECout <i>Real and imaginary part of compressed spectrum, as two input channels for encoder.</i>		MSE + Complex MSE <i>MSE of magnitude spectra, and MSE of complex spectra (real and imaginary parts).</i>	None	/	C	Y



Summary of AEC

	Input feature	Network architecture	Loss function	Auxiliary Task	Training data	Real or Complex	Hybrid or not
GTCNN	Mic / Ref <i>Real and imaginary part of compressed spectrum stacked together.</i>	Complex inputs → Encoder → Sequence Modeling → Decoder → Real/Imag Speaker embedding → Sequence Modeling FFT/Conv/CCConv/GConv (under Encoder) F-T-LSTM/TCN/GTCN/DPRNN/DPT (under Sequence Modeling and Decoder)	Complex MSE <i>Real and imaginary parts optimized at the same time.</i>	Fast ResNet <i>A lightweight SRE neural network.</i>	200h	C	N



声学回声消除技术最新进展

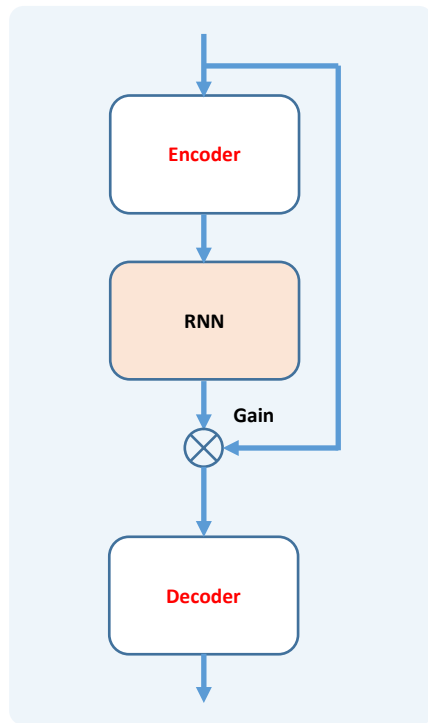


噪声抑制技术最新进展

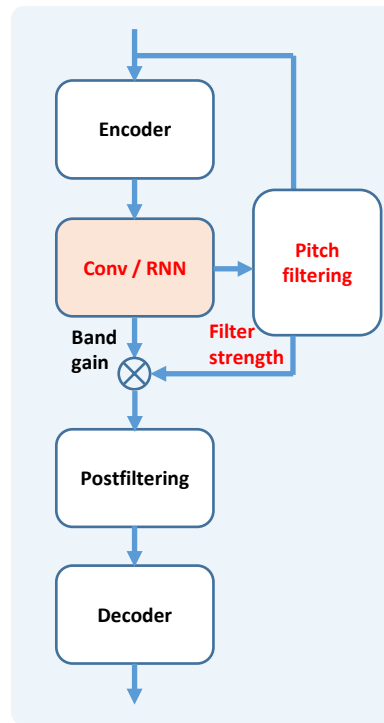


RNNoise & PercepNet

RNNoise

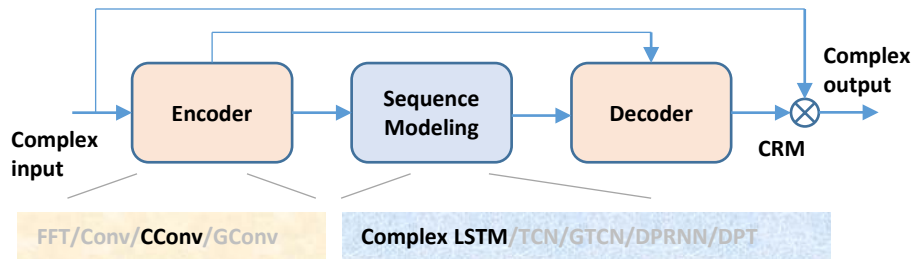


PercepNet



Deep Complex Convolution Recurrent Network

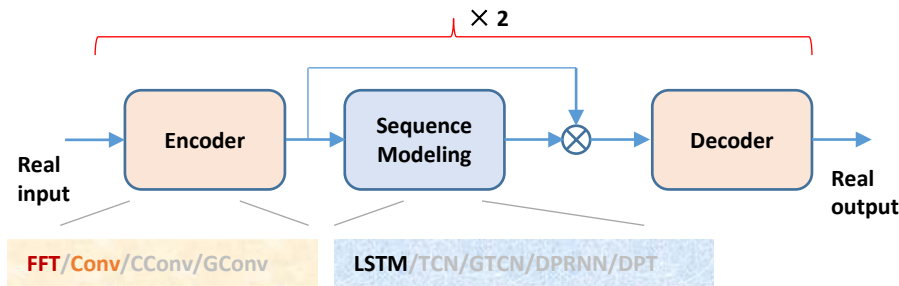
DCCRN:



- Encoder: **Complex** Conv
- Sequence Modeling: **Complex** LSTM

Dual-Signal Transformation LSTM Network

DTLN:



Stacked architecture:

第一阶段:

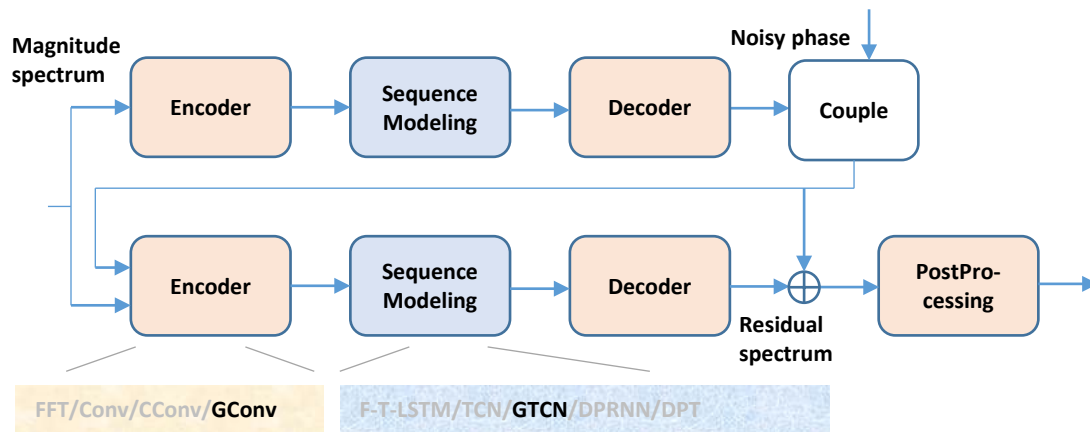
- 时频域网络, 预测time-frequency mask;
- 只考虑幅度信息, 不考虑相位信息。

第二阶段:

- 时域网络, 预测latent space mask;
- 同时考虑幅度和相位。

Two-Stage Complex Network with Post-Processing

TSCN-PP:

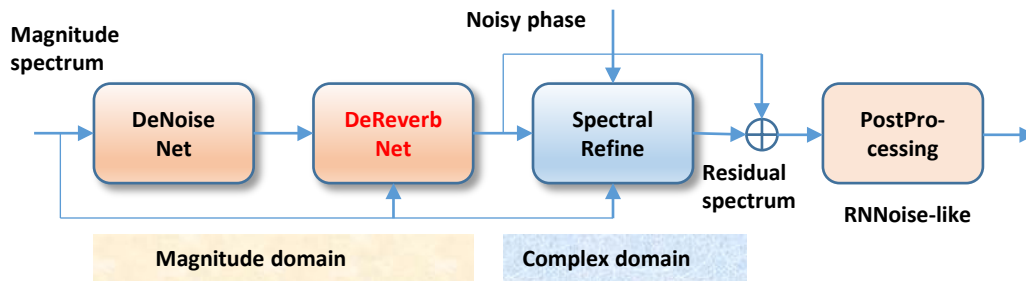


- Stage1: 只考虑幅度信息
- Stage2: 同时考虑幅度和相位信息
- Post-Processing: RNNnoise-like



Simultaneous Denoising and Dereverberation Network

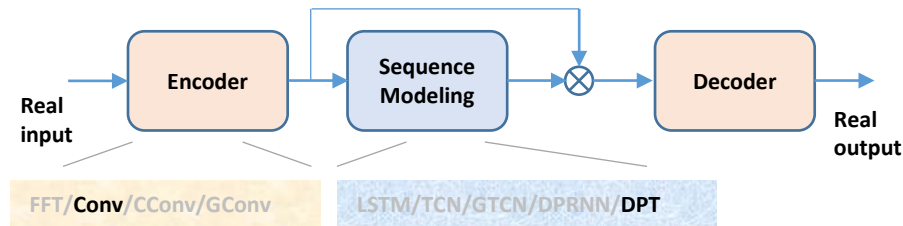
SDD-Net:



- 在幅度域处理降噪和解混响
- 在复数域做refinement
- Post-Processing: RNNnoise-like

Transformer-based Neural Network for Separation

SepFormer

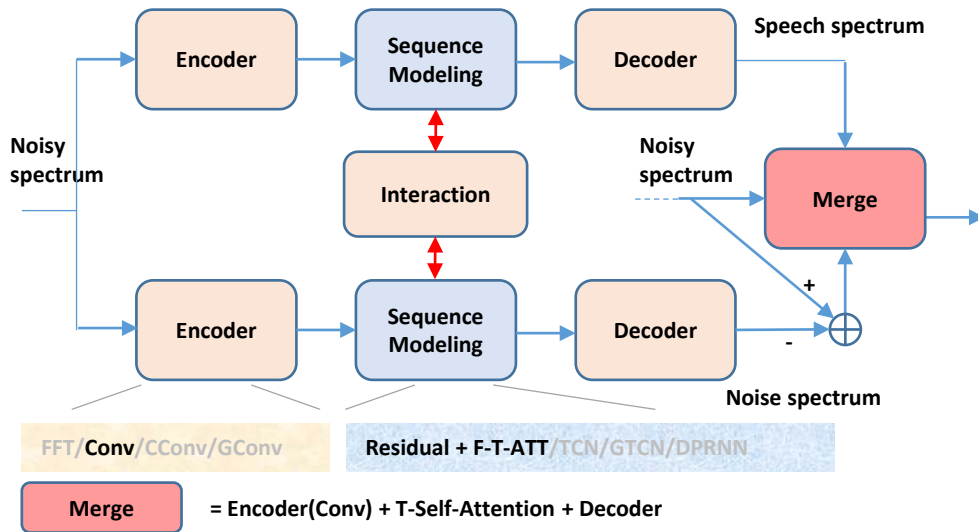


- 基于时域的语音分离网络
- Sequence Modeling采用了Dual-Path Transformer



Speech and Noise Modeling

SN-Net

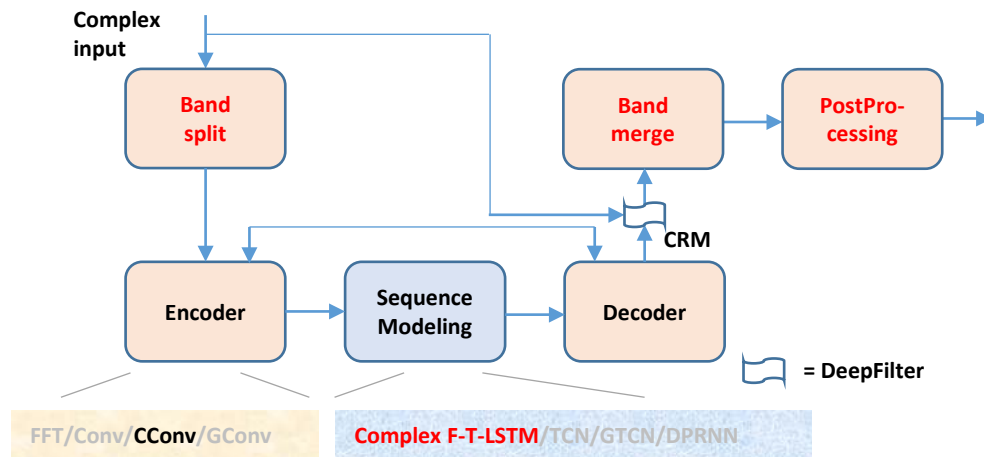


- 两个 “E-S-D” 架构同时估计语音和噪声谱
- Merge: 也是一个 “E-S-D” 架构



Channel-wise Subband DCCRN

DCCRN+

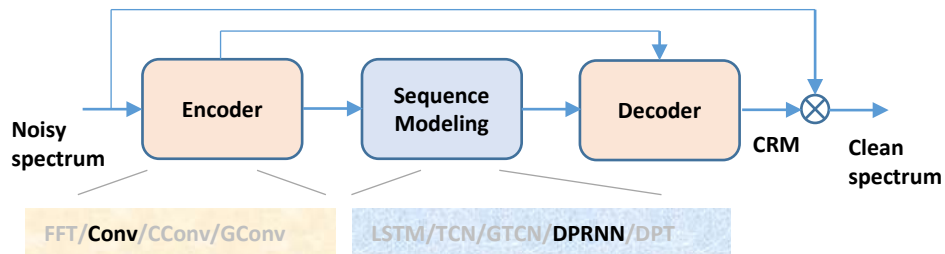


- 将DCCRN扩展到子带中：
 - 降低运算量和模型大小
 - 语音信号不同频率（子带）内的pattern可能不一样，分解成子带更容易建模
- Complex LSTM改进成Complex F-T-LSTM

DCCRN+: Channel-wise Subband DCCRN with SNR Estimation for Speech Enhancement. Shubo Lv, etc. 2021

Dual-Path Convolution Recurrent Network

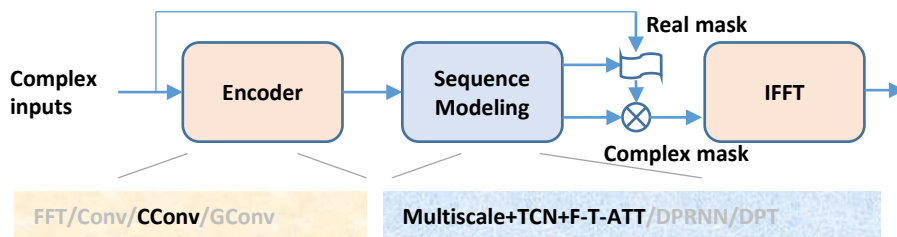
DPCRN



- DPRNN作为序列建模网络

Multi-scale Temporal Frequency Convolutional Network with Axial Attention

MTFAA-Net

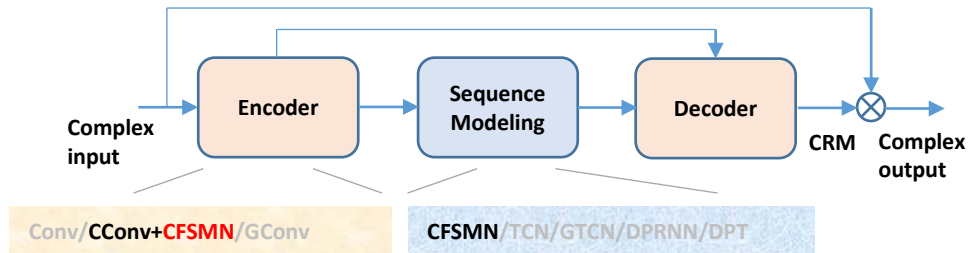


- 多尺度特征
- Temporal Frequency Model:
 - 序列建模: TCN with Conv2d
 - Axial Attention = F-T-Attention
- 两阶段的mask估计



Frequency Recurrent CRN

FRCRN

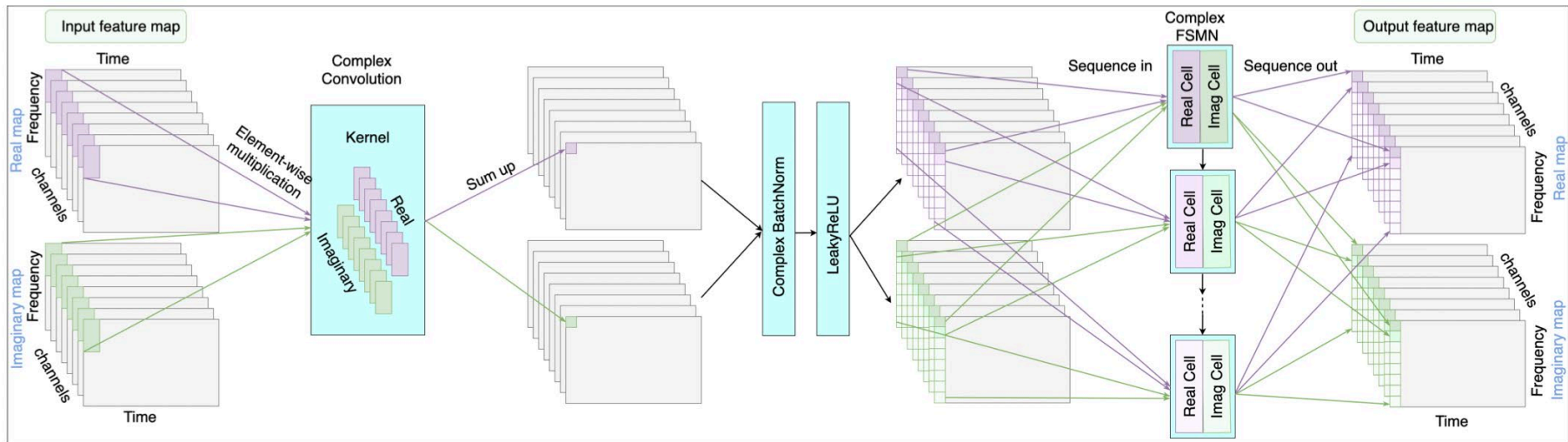


- 将Frequency Recurrence提前到Encoder阶段，提升特征的表征能力。



Frequency Recurrent CRN

CR (Convolutional Recurrent) block in FRCRN:



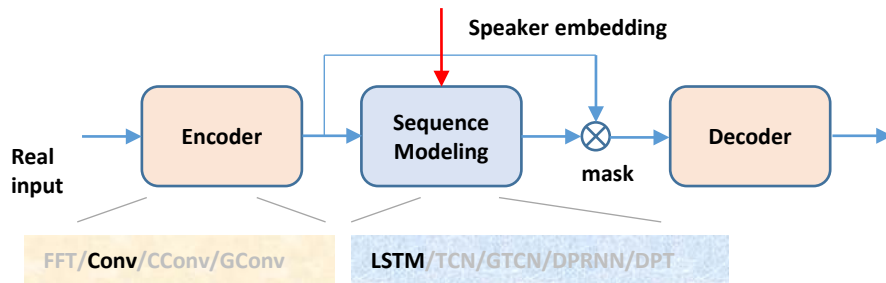
Complex Conv2d

Complex FSMN



End-to-End Enhancement Network

E3Net

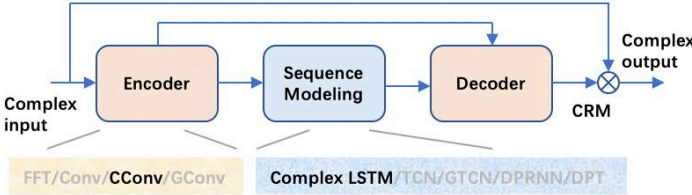
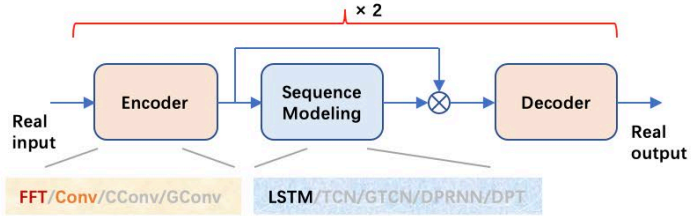


- 个性化语音增强
- E3Net + Knowledge Distillation + Multi-task
 - 可以利用更多大量的无标注数据
 - 可以应用到任何降噪网络中

Fast Real-time Personalized Speech Enhancement: End-to-End Enhancement Network (E3Net) and Knowledge Distillation. M Thakker, etc. 2022



Summary of NS

	Input feature	Network architecture	Loss function	Auxiliary Task	Training data	Real or Complex	Hybrid or not
DCCRN	Complex-value STFT of noisy speech		SI-SNR	None	IS2020 DNS dataset: 500h speech + 180h noise	C	N
DTLN	Time-domain noisy speech		Negative SNR	None	IS2020 DNS dataset: 500h	R	N

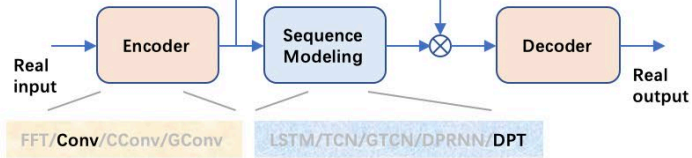
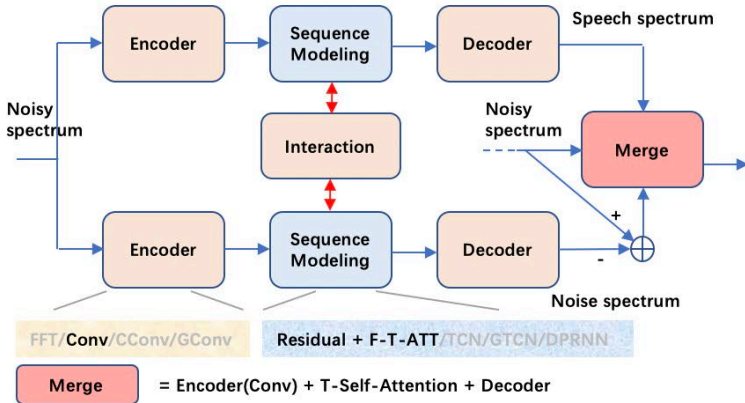


Summary of NS

Input feature	Network architecture	Loss function	Auxiliary Task	Training data	Real or Complex	Hybrid or not
TSCN-PP Spectrum of Noisy Speech		MSE Stage1: MSE of magnitude spectrum; Stage 2: MSE of real and imaginary part + MSE of magnitude + Stage1 loss.	None	517h	R+C	Y
SDD-Net Spectrum of Noisy Speech		MSE	None	~500h	R+C	Y



Summary of NS

	Input feature	Network architecture	Loss function	Auxiliary Task	Training data	Real or Complex	Hybrid or not
Sep-former	Time-domain Mixture Signal		SI-SNR	None	WSJ0-2mix 30h	R	N
SN-Net	Spectrum of Noisy Speech <i>With real and imaginary part as two input channels.</i>		MSE <i>MSE on power-law compressed STFT spectrum, in THREE BRANCHE: Speech, Noise, Merge branch.</i>	None	500h+	C	N



Summary of NS

	Input feature	Network architecture	Loss function	Auxiliary Task	Training data	Real or Complex	Hybrid or not
DCCRN+	Complex-value STFT of noisy speech	Complex input Band split Encoder Sequence Modeling Decoder Band merge PostProcessing CRM = DeepFilter FFT/Conv/CConv/GConv Complex F-T-LSTM/TCN/GTCN/DPRNN	SI-SNR + MSE (of SNR) <i>MSE on estimated and real SNR.</i>	Multi-task <i>SNR Estimation.</i>	760h	C	N
DPCRNN	Spectrum of Noisy Speech	Noisy spectrum Encoder Sequence Modeling Decoder CRM Clean spectrum FFT/Conv/CConv/GConv LSTM/TCN/GTCN/DPRNN/DPT	Negative SNR + MSE <i>MSE on real part, imaginary part, and magnitude.</i>	None	500h+	C	N



Summary of NS

	Input feature	Network architecture	Loss function	Auxiliary Task	Training data	Real or Complex	Hybrid or not
MTFAA-Net	Complex-value STFT of noisy speech		PLCPA <i>MSE on power-law compressed spectrum, i.e. mag + phase loss.</i>	None	/	C	Y
FRCRN	Complex-value STFT of noisy speech		SI-SNR + MSE <i>MSE of real part and imaginary part of CRM.</i>	None	3000h	C	Y



Summary of NS

	Input feature	Network architecture	Loss function	Auxiliary Task	Training data	Real or Complex	Hybrid or not
E3Net	Time domain noisy speech	Real input → Encoder (FFT/Conv/CConv/GConv) → Sequence Modeling (LSTM/TCN/GTCN/DPRNN/DPT) → mask → Decoder Speaker embedding is added to Sequence Modeling.	SE Loss + ASR Loss	SRE <i>Speaker embedding.</i> ASR <i>Multi-task training with ASR.</i>	2000h + 1302h ASR Data	R	N
PoCoNet	Time domain noisy speech	Real input → STFT → Sequence Modeling (Dense U-net with Self-attention) → CRM → ISTFT Frequency-Positional Encoding is added to Sequence Modeling.	L1 Loss <i>Biased L1 loss on magnitude + L1 loss on waveform domain.</i>	Semi-Supervised	600h supervised + 800h Semi-supervised	C	Y



结语

感谢各位聆听!

Thanks for Listening